

Cotangent Message Passing for Exact ReLU Feasibility and the Existential Theory of the Reals

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Abstract

We study a decision formulation of exact training for finite feed-forward rectified-linear-unit (ReLU) networks. The trainable parameters are shared across all samples, whereas preactivations, activations, and slack variables are sample-local. This induces a block-angular polynomial system and an iterated tensor-product description of its coordinate ring over the shared parameter algebra. We establish the corresponding decomposition of relative Kähler differentials, formulate an intrinsic cotangent separator message as the image of the kernel of an internal Jacobian block, and prove its exact factorization property for the associated linearized system. These results hold without selecting an activation pattern; the exact ReLU graph, including its switching boundary, is represented by polynomial equalities with existential square slacks.

The central question is whether these cotangent messages decide the original zeroth-order feasibility problem. We give an in-scope one-ReLU counterexample: for every positive target, the evaluated scalar cotangent message is the full separator space, even though the exact surviving parameter set is the singleton containing that target. Hence no decoder whose only semantic input is the first-order message subspace can decide all exact ReLU instances. This separates the proved linear separator theorem from the global nonlinear decision invariant required for a polynomial-time algorithm. We record the precise additional obligations—a complete zeroth-order message, a polynomial-size representation theorem, a Turing-model bit bound, and a model-matching reduction from the existential theory of the reals. We also state the resulting class collapse conditionally. In particular, this draft does not claim an unconditional proof that $P = \exists\mathbb{R}$ or $P = NP$.

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1 Introduction

Exact feasibility questions for neural networks sit at the intersection of real algebraic geometry, computational complexity, and formal verification. Once approximation and numerical optimization are removed, a finite feed-forward ReLU network with unknown real weights and rational data defines a finite system of polynomial equalities and inequalities. The natural decision question is whether that system has a real solution. Closely related neural network training problems are known to be complete for the complexity class $\exists\mathbb{R}$ [AKM21]; this class contains NP and is contained in PSPACE [Can88, SCM24].

This paper investigates whether the repeated-sample structure of exact neural training admits an exact separator algorithm. A single parameter vector θ is shared across the dataset, while each sample has its own preactivations, activations, and auxiliary variables. At the level of polynomial equations this yields a block-angular Jacobian. At the level of commutative algebra it yields an iterated tensor product over the shared parameter algebra. These facts motivate *Cotangent Message Passing* (CMP): eliminate a local block from the row space of a linearized constraint system and retain only the induced subspace on a separator.

The distinction between a linearized relation and an exact existential projection is decisive. The cotangent module records first-order behavior. The decision problem asks whether the semialgebraic set is empty. Neither the universal property of Kähler differentials nor a pointwise Jacobian identity identifies these two objects. A proof of an exact CMP decider therefore needs an additional semantic invariant:

$$s \in \text{Msg}_v \iff \exists x_{\text{int}(T_v)} F_{T_v}(x, s). \quad (1)$$

It must also prove that the representation of Msg_v , its merge operation, and its projection operation remain polynomially bounded.

1.1 Contributions

The present draft makes the following claims, each with an explicit proof status.

1. We define EXACT-RELU-NNTP-DECISION as a Boolean feasibility language with rational input and unrestricted real witnesses. “Training” describes the origin of the constraints; the output is a Boolean, not a weight vector.
2. We prove an exact complementarity description of a ReLU gate and its existential square-slack polynomialization. The description includes the switching point $z = a = 0$.
3. We derive the shared-base tensor-product description of the equality coordinate ring and the direct-sum decomposition of relative Kähler differentials after base change. We then display the induced block-angular Jacobian.
4. We define the intrinsic first-order CMP message $\mathcal{E}(A \rightarrow S) = J_{AS}(\ker J_A)$ and prove exact separator factorization for the associated linear row-space problem. This theorem is independent of a null-space basis.
5. We prove that the first-order message is not a complete invariant of exact ReLU feasibility. A one-ReLU family with positive target y has exact parameter projection $\{y\}$, but its evaluated scalar CMP message equals $\mathbb{R}$ for every $y > 0$. Thus targets one and two yield identical messages and different exact feasibility at $\theta = 1$.
6. We isolate the remaining obligations for the proposed complexity collapse: a polynomially represented zeroth-order message, a correct global decider, polynomial Turing-model bit complexity, and a reduction landing in the exact language defined here. We formalize the final class-theoretic implication conditionally.

1.2 Status of the proposed collapse

The intended implication has the form

$$\exists \mathbb{R} \leq_p \text{EXACT-RELU-NNTP-DECISION} \in P \implies P = NP = \exists \mathbb{R}. \quad (2)$$

The implication in [equation \(2\)](#) is standard. The right-hand premise is not supplied by the first-order CMP theorem. Indeed, [section 10](#) disproves the required completeness statement for a decoder that receives only the existing cotangent message subspace. The collapse is therefore recorded as a target conditional theorem, not as an unconditional result.

This separation is more than a matter of exposition. The counterexample is inside the stated problem class, uses a single ReLU, and occurs at a positive satisfying point. It cannot be repaired by excluding ReLU switching singularities. Any successful exact message must preserve zeroth-order target constants and global component information that differentiation can erase.

1.3 Organization

[Sections 2 to 4](#) define the computational and algebraic setting. [Sections 5 to 7](#) establish the shared-base structure. [Sections 8 and 9](#) develop the separator decomposition and the proved linear CMP theorem. [Section 10](#) gives the semantic projection invariant and the counterexample to cotangent-only completeness. The algorithmic and complexity consequences are audited in [sections 11 and 12](#). [Sections 13 and 14](#) state the exact reduction and collapse obligations. Boundary behavior, related work, and possible repair directions appear in the remaining sections.

2 Background and preliminaries

2.1 The existential theory of the reals

An existential sentence over the reals has the form

$$\exists x_1, \dots, x_m \in \mathbb{R} : \Psi(x_1, \dots, x_m), \quad (3)$$

where Ψ is a quantifier-free Boolean combination of polynomial sign conditions with integer coefficients. The decision problem ETR asks whether such a sentence is true. The class $\exists\mathbb{R}$ is the closure of ETR under polynomial-time many-one reductions. Standard containments give

$$P \subseteq NP \subseteq \exists\mathbb{R} \subseteq PSPACE. \quad (4)$$

We use the discrete Turing model: coefficients and combinatorial structure are encoded by finite binary strings, while quantified witnesses range over the reals. Surveys and algorithms for real feasibility may be found in [\[BPR06, Ren92, SCM24\]](#); Canny’s algorithm supplies the PSPACE upper bound [\[Can88\]](#).

Definition 2.1 (Semialgebraic set). *A subset $X \subseteq \mathbb{R}^d$ is semialgebraic if it is defined by a finite Boolean combination of polynomial equalities and inequalities with real coefficients. A basic semialgebraic set is defined by a conjunction of such sign conditions.*

Existentially quantified square variables can replace weak nonnegativity:

$$t \geq 0 \iff \exists u \in \mathbb{R} : t = u^2. \quad (5)$$

This device permits an equality-only ambient variety whose real projection is the intended basic semialgebraic set. The projection operation remains part of the semantics; the equality coordinate ring alone does not remember which variables were introduced existentially.

2.2 Coordinate rings and differentials

Let k be a commutative base field and B a finitely generated k -algebra. The module of Kähler differentials $\Omega_{B/k}$ is the B -module equipped with a universal k -derivation $d : B \rightarrow \Omega_{B/k}$. For a tower $k \rightarrow A \rightarrow B$, there is a right-exact transitivity sequence

$$B \otimes_A \Omega_{A/k} \longrightarrow \Omega_{B/k} \longrightarrow \Omega_{B/A} \longrightarrow 0. \quad (6)$$

These standard constructions are reviewed, for example, in [Eis95, The].

If

$$B = k[x_1, \dots, x_n]/(f_1, \dots, f_m), \quad (7)$$

then $\Omega_{B/k}$ admits the presentation

$$B^m \xrightarrow{J_F^T} B^n \longrightarrow \Omega_{B/k} \longrightarrow 0, \quad (8)$$

where $J_F = (\partial f_i / \partial x_j)$. This is a module over the entire quotient ring. Evaluating at a real point p tensors the presentation with the residue field at p and produces pointwise cotangent information. The symbolic module and any one evaluated vector space are distinct objects.

2.3 Linear maps and intrinsic messages

Let $J_A : L_A \rightarrow X_A$ and $J_{AS} : L_A \rightarrow X_S$ be linear maps over a field. The intrinsic separator message is

$$\mathcal{E}(A \rightarrow S) := J_{AS}(\ker J_A) \subseteq X_S. \quad (9)$$

If the maps are represented by row blocks and $P_A^\perp$ has rows spanning the relevant left null space, then a matrix representing the same subspace is

$$M_{A \rightarrow S} = P_A^\perp J_{AS}. \quad (10)$$

The subspace, rather than the chosen basis matrix, is the invariant.

2.4 Tree decompositions

A tree decomposition of a graph $G = (V, E)$ is a tree whose nodes carry bags $B_t \subseteq V$ such that every vertex appears in a connected family of bags and every edge is contained in some bag. Its width is $\max_t |B_t| - 1$; the treewidth $\text{tw}(G)$ is the minimum width. Bounded treewidth often enables exact dynamic programming when each separator message has a controlled representation [Bod96]. Treewidth alone does not bound the descriptive complexity of projected semialgebraic sets.

2.5 Proof-status vocabulary

To prevent a conditional dependency from being silently promoted, we use four labels in this draft:

Proved	A complete mathematical proof is present or a cited standard result applies with its hypotheses checked.
Conditional	The implication is proved assuming explicitly named certificates or lemmas.
Open	The statement is required by the target theorem but is not presently derived.
Refuted	A counterexample disproves the statement as formulated.

3 The exact ReLU NNTP decision problem

We define a finite Boolean language. The definition is deliberately separated from any optimization method.

Definition 3.1 (Network architecture). *A feed-forward ReLU architecture is a finite directed acyclic graph with input, affine, ReLU, and output nodes. An affine node computes a rational affine combination of earlier activations in which a designated subset of coefficients are coordinates of a shared parameter vector $\theta \in \mathbb{R}^P$. Products of an unknown weight and a sample-local activation are polynomial terms. A ReLU node maps a preactivation z to $a = \max\{0, z\}$.*

For each sample $i \in [N]$, write $S^{(i)} \in \mathbb{R}^{K_i}$ for every sample-local preactivation, activation, output, and auxiliary slack variable. The parameters θ are not copied by sample. Rational inputs and targets are substituted as constants.

Definition 3.2 (Exact-ReLU-NNTP instance). *An instance I consists of:*

1. *a finite feed-forward architecture and a designation of trainable parameters;*
2. *a finite rational dataset $(x_i, y_i)_{i=1}^N$;*
3. *exact affine-transition constraints, exact ReLU constraints, and exact output-target equalities for every sample; and*
4. *a binary encoding of all rational coefficients and incidence data.*

Its feasibility predicate is

$$\text{Feas}(I); :\iff \exists \theta \in \mathbb{R}^P \exists S^{(1)} \in \mathbb{R}^{K_1} \dots \exists S^{(N)} \in \mathbb{R}^{K_N} : \bigwedge_{i=1}^N \Phi_i(\theta, S^{(i)}). \quad (11)$$

Definition 3.3 (EXACT-RELU-NNTP-DECISION). *Given the binary encoding of I , return YES if and only if $\text{Feas}(I)$ holds.*

No witness is required as output. The definition does not mention a loss surface, gradient descent, a local optimum, or numerical tolerance. Exact target equality is used in this draft; a threshold-loss variant is a different formal language unless a model-matching reduction is supplied.

3.1 Input size and uniformity

Let L be the maximum coefficient bit length, and let $n = |\langle I \rangle|$ be the total binary encoding length. The encoding explicitly lists the architecture, trainable parameter incidences, samples, and rational constants. Consequently

$$N \leq n, \quad P \leq n, \quad K_i \leq n \quad (12)$$

for any reasonable noncompressed encoding. These inequalities do not by themselves prove that an algebraic algorithm runs in polynomial time: the size of intermediate polynomials, formulas, or algebraic numbers must also be controlled.

3.2 Sample locality

The constraint family has the syntactic form

$$\Phi_I(\theta, S^{(1)}, \dots, S^{(N)}) = \bigwedge_{i=1}^N \Phi_i(\theta, S^{(i)}). \quad (13)$$

Thus distinct samples share θ and no sample-local variable. This property drives both the tensor decomposition and the star-shaped dependency graph below.

Proposition 3.4 (Semantic parameter projection). *For each sample define*

$$Q_i := \{\theta \in \mathbb{R}^P : \exists S^{(i)} \text{ } \Phi_i(\theta, S^{(i)})\}. \quad (14)$$

Then

$$\text{Feas}(I) \iff \bigcap_{i=1}^N Q_i \neq \emptyset. \quad (15)$$

Proof. If a common assignment θ and local witnesses satisfy [equation \(11\)](#), then $\theta \in Q_i$ for every i . Conversely, if $\theta \in \bigcap_i Q_i$, choose a local witness for each of the finitely many samples. These witnesses share no local variables and therefore combine with θ into a global witness. $\square$

[Proposition 3.4](#) is an exact zeroth-order separator theorem. It does not give an efficient representation of the sets Q_i or an efficient method for testing their intersection.

4 Exact algebraic and semialgebraic ReLU encoding

The graph of a ReLU is basic semialgebraic and has a particularly small complementarity description.

Lemma 4.1 (Exact ReLU complementarity). *For $z, a \in \mathbb{R}$,*

$$a = \text{ReLU}(z) \iff a \geq 0, \quad a - z \geq 0, \quad a(a - z) = 0. \quad (16)$$

Proof. Suppose first that $a = \max\{0, z\}$. If $z \leq 0$, then $a = 0$, so $a \geq 0$, $a - z = -z \geq 0$, and the product vanishes. If $z \geq 0$, then $a = z$, so both weak inequalities hold and $a - z = 0$.

Conversely, the product equation implies either $a = 0$ or $a - z = 0$. In the first case, $a - z \geq 0$ implies $z \leq 0$, whence $a = 0 = \max\{0, z\}$. In the second case, $a = z$ and $a \geq 0$, whence $a = z = \max\{0, z\}$. $\square$

The three regimes are all present:

$$z < 0 \implies a = 0, \quad (17)$$

$$z = 0 \implies a = 0, \quad (18)$$

$$z > 0 \implies a = z. \quad (19)$$

No disequality $z \neq 0$ is introduced.

Lemma 4.2 (Equality-only polynomialization). *For $z, a \in \mathbb{R}$,*

$$a = \text{ReLU}(z) \iff \exists u, v \in \mathbb{R} : a = u^2, \quad a - z = v^2, \quad a(a - z) = 0. \quad (20)$$

Proof. By lemma 4.1, a ReLU assignment makes a and $a - z$ nonnegative. Each is therefore a square in $\mathbb{R}$. Conversely, the two square equalities imply the weak inequalities, and the complementarity equation then allows an application of lemma 4.1. $\square$

Remark 4.3 (Projection is part of the semantics). The equality variety in variables (z, a, u, v) is not literally the graph in the (z, a) -plane. Its projection to (z, a) is the graph. The slack variables therefore belong to the existential local state.

Proposition 4.4 (Membership in $\exists\mathbb{R}$). *The language EXACT-RELU-NNTP-DECISION belongs to $\exists\mathbb{R}$.*

Proof. Introduce real variables for the shared parameters, every sample-local preactivation and activation, and the two square slacks of each ReLU. Affine transitions with trainable weights are polynomial equalities of degree at most two in the unknowns. Exact targets are polynomial equalities. Replace every ReLU by equation (20). The number of variables and constraints and the coefficient encoding length grow polynomially with the explicit instance encoding. The resulting existential conjunction of polynomial equalities is true exactly when the instance is feasible. $\square$

4.1 The switching boundary and singularity

The complementarity polynomial

$$g(z, a) = a(a - z) \quad (21)$$

has gradient

$$\nabla g(z, a) = (-a, 2a - z). \quad (22)$$

At $(z, a) = (0, 0)$, this gradient vanishes. Thus a Jacobian-based regularity argument cannot cover the switching point merely because the global Kähler module exists there. Kähler differentials are defined at singular points, but their fibers can change rank and acquire information not present on a smooth stratum. The exact semantics above remains valid at the boundary; only the first-order regularity changes.

5 Shared-parameter coordinate-ring decomposition

We now retain the polynomial equalities, including square-slack equations, and separate the one shared parameter algebra from sample-local algebras. Let $k = \mathbb{R}$ for the geometric statements and put

$$A = k[\theta_1, \dots, \theta_P]. \quad (23)$$

For sample i , let $S^{(i)}$ denote a finite tuple of local indeterminates and define

$$B_i = A[S^{(i)}], \quad \mathcal{R}^{(i)} = B_i / I^{(i)}, \quad (24)$$

where $I^{(i)}$ is generated by the polynomial equalities belonging to that sample. In the common ambient polynomial algebra

$$R = A[S^{(1)}, \dots, S^{(N)}], \quad (25)$$

each $I^{(i)}$ is understood through extension from B_i to R , and

$$I = \sum_{i=1}^N I^{(i)} R, \quad \mathcal{R} = R / I. \quad (26)$$

Proposition 5.1 (Ambient shared-base tensor product). *There is a canonical A -algebra isomorphism*

$$A[S^{(1)}, \dots, S^{(N)}] \cong A[S^{(1)}] \otimes_A \cdots \otimes_A A[S^{(N)}]. \quad (27)$$

Proof. Both sides are the coproduct in commutative A -algebras of the polynomial A -algebras on the disjoint variable sets $S^{(i)}$. Concretely, the map sends each variable to its copy in the corresponding tensor factor. The inverse multiplies the images of pure tensors in the common polynomial ring. The two maps agree on A and on every variable, and hence are inverse by the universal property of polynomial algebras. $\square$

Theorem 5.2 (Quotient tensor decomposition). *With the notation above, there is a canonical A -algebra isomorphism*

$$\boxed{\mathcal{R} \cong \mathcal{R}^{(1)} \otimes_A \cdots \otimes_A \mathcal{R}^{(N)}}. \quad (28)$$

Proof. It suffices to treat two factors and iterate. The canonical bilinear map $B_1 \times B_2 \rightarrow (B_1 \otimes_A B_2)/J$, where J is generated by the images of $I^{(1)} \otimes B_2$ and $B_1 \otimes I^{(2)}$, vanishes whenever either input belongs to the corresponding ideal. It therefore descends to a map

$$(B_1/I^{(1)}) \otimes_A (B_2/I^{(2)}) \longrightarrow (B_1 \otimes_A B_2)/J. \quad (29)$$

The universal property gives an inverse, so the map is an isomorphism. Under [equation \(27\)](#), the ideal J corresponds to the sum of the two extended sample ideals. Iteration gives [equation \(28\)](#). $\square$

Remark 5.3 (One parameter algebra). The tensor product is over A . It identifies the parameter coordinates in all local factors. A tensor product over k would create independent copies of the parameters and would model a different feasibility problem.

5.1 Equality rings versus feasible real sets

[Theorem 5.2](#) concerns polynomial equalities. The exact decision problem concerns real points together with the existential interpretation of slack variables. Passing to a quotient ring does not perform existential projection, choose a real connected component, or decide whether real points exist. These distinctions matter because two systems can have closely related differential modules while having different real feasibility.

The global real feasible set may be written

$$X_I(\mathbb{R}) = \left\{ (\theta, S^{(1)}, \dots, S^{(N)}) : F^{(i)}(\theta, S^{(i)}) = 0 \text{ for all } i \right\}, \quad (30)$$

after square-slack polynomialization. The accepted parameter set is its projection to θ , equivalently the intersection in [equation \(15\)](#). The coordinate-ring decomposition explains the sample incidence pattern, but it does not compute this projection.

6 Kähler differential structure

The shared-base tensor product has a corresponding decomposition of relative differentials.

Theorem 6.1 (Relative differential decomposition). *Let $\mathcal{R}$ be the iterated tensor product in [equation \(28\)](#). There is a canonical $\mathcal{R}$ -module isomorphism*

$$\boxed{\Omega_{\mathcal{R}/A} \cong \bigoplus_{i=1}^N \left(\Omega_{\mathcal{R}^{(i)}/A} \otimes_{\mathcal{R}^{(i)}} \mathcal{R} \right)}. \quad (31)$$

Proof. For two A -algebras B, C , an A -derivation from $B \otimes_A C$ to a $(B \otimes_A C)$ -module M is uniquely determined by its restrictions to B and C . Conversely, two such derivations define $d(b \otimes c) = c d_B(b) + b d_C(c)$. The universal property of Kähler differentials therefore gives

$$\Omega_{(B \otimes_A C)/A} \cong (\Omega_{B/A} \otimes_B (B \otimes_A C)) \oplus (\Omega_{C/A} \otimes_C (B \otimes_A C)). \quad (32)$$

Iterating this isomorphism and applying [theorem 5.2](#) proves the claim. $\square$

This theorem is global and requires no smoothness hypothesis. Its meaning is relative: after the shared parameter algebra is held fixed, different samples have independent differential generators and relations.

Proposition 6.2 (Absolute-to-relative transitivity). *For $k \rightarrow A \rightarrow \mathcal{R}$, there is a right-exact sequence*

$$\mathcal{R} \otimes_A \Omega_{A/k} \longrightarrow \Omega_{\mathcal{R}/k} \longrightarrow \Omega_{\mathcal{R}/A} \longrightarrow 0. \quad (33)$$

Proof. This is the standard Jacobi–Zariski transitivity sequence for Kähler differentials [[The](#)]. The first map sends a parameter differential from A to its image in the global algebra; the second quotient forgets parameter motion. $\square$

The first term in [equation \(33\)](#) is precisely the channel through which sample-relative pieces couple in the absolute module. Right exactness does not assert that the first map is injective, nor that the sequence splits.

6.1 Jacobian presentation

Choose polynomial equality generators $F = (F_1, \dots, F_m)$ for I in variables $x = (\theta, S^{(1)}, \dots, S^{(N)})$. Then

$$\mathcal{R}^m \xrightarrow{J_F^\top} \mathcal{R}^{P+\sum_i K_i} \longrightarrow \Omega_{\mathcal{R}/k} \longrightarrow 0. \quad (34)$$

The image of $J_F^\top$ records differential relations $dF_j = 0$. Different generating families can give different presentation matrices while presenting isomorphic modules.

Remark 6.3 (Global module versus fiber). For a real solution p , tensoring [equation \(34\)](#) with the residue field at p evaluates all coefficients. The resulting vector-space cokernel describes a cotangent fiber. Equality of such fibers, or equality of their row spaces, need not imply equality of the original ideals or equality of their real projections.

Remark 6.4 (Singular points). The module $\Omega_{\mathcal{R}/k}$ and the exact sequence [equation \(33\)](#) exist at singular points. What fails there is a naive identification of module rank with the dimension of a smooth manifold. Our algebraic decompositions remain valid; claims based on constant-rank linear algebra require separate hypotheses.

7 Block-angular Jacobian structure

Order the variables as

$$(\theta, S^{(1)}, \dots, S^{(N)}) \quad (35)$$

and the polynomial equations sample by sample. Write

$$C_i = \frac{\partial F^{(i)}}{\partial \theta}, \quad D_i = \frac{\partial F^{(i)}}{\partial S^{(i)}}. \quad (36)$$

Lemma 7.1 (Cross-sample derivatives vanish). *If $i \neq j$, then*

$$\frac{\partial F^{(i)}}{\partial S^{(j)}} = 0. \quad (37)$$

Proof. By sample locality, no indeterminate in $S^{(j)}$ occurs in any polynomial of $F^{(i)}$. A formal partial derivative with respect to an absent indeterminate is zero. $\square$

Theorem 7.2 (Block-angular presentation). *With the order above, the global Jacobian is*

$$J_F = \begin{bmatrix} C_1 & D_1 & 0 & \cdots & 0 \\ C_2 & 0 & D_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ C_N & 0 & 0 & \cdots & D_N \end{bmatrix}. \quad (38)$$

Proof. The parameter block can be nonzero in every sample because parameters are shared. The local diagonal blocks contain derivatives with respect to that sample's state. Every off-diagonal local block is zero by [lemma 7.1](#). $\square$

The theorem holds over the polynomial quotient ring before evaluation and over $\mathbb{R}$ after evaluation at any real point. These are different coefficient structures. Gaussian elimination is immediately available for a numerical matrix over a field. Symbolic elimination over a multivariate quotient ring requires additional algebra and cannot be charged as ordinary field row reduction without justification.

7.1 Pointwise tangent compatibility

At a real solution p , a first-order perturbation $(\delta\theta, \delta S^{(1)}, \dots, \delta S^{(N)})$ satisfies the linearized equations exactly when

$$C_i(p)\delta\theta + D_i(p)\delta S^{(i)} = 0 \quad (i \in [N]). \quad (39)$$

Eliminating $\delta S^{(i)}$ gives a constraint on $\delta\theta$. This is a tangent-extension question at a known point. It does not ask whether a point p exists, and it does not recover the constant terms of the original equations.

Proposition 7.3 (Local linearized extension criterion). *Over a field, let $P_i^\perp$ have row space $\ker D_i(p)^\top$. Then [equation \(39\)](#) has a solution $\delta S^{(i)}$ if and only if*

$$P_i^\perp C_i(p)\delta\theta = 0. \quad (40)$$

Proof. The vector $-C_i(p)\delta\theta$ belongs to the image of $D_i(p)$ exactly when it is annihilated by every element of the left null space of $D_i(p)$. Writing a basis of that null space as the rows of $P_i^\perp$ gives [equation \(40\)](#). $\square$

Stacking the reduced constraints yields the root matrix used by differential CMP. The criterion is exact for the linear system at p .

8 Dependency topology and separators

Construct a primal dependency graph whose vertices are scalar variables and in which two variables are adjacent when they occur together in an input constraint. A coarse tree decomposition is obtained from the bags

$$B_i = \{\theta_1, \dots, \theta_P\} \cup S^{(i)} \quad (i \in [N]), \quad (41)$$

connected through a central parameter bag

$$B_0 = \{\theta_1, \dots, \theta_P\}. \quad (42)$$

Proposition 8.1 (Star decomposition). *The bags $B_0, B_1, \dots, B_N$, with every B_i adjacent to B_0 , form a tree decomposition of the coarse sample-local dependency graph.*

Proof. Every constraint belongs to one sample and therefore all of its variables lie in the corresponding bag B_i . Each local variable occurs in only that bag. Each parameter occurs in the center and in every sample bag, which is a connected subtree. Hence the edge-cover and running-intersection conditions hold. $\square$

Corollary 8.2 (Width bound). *If $K = \max_i K_i$, then*

$$\boxed{\text{tw}(G) \leq P + K - 1.} \quad (43)$$

Proof. The largest bag in equation (41) has at most $P + K$ vertices. $\square$

Increasing N adds bags but does not enlarge their shared intersection. This fact explains why sample-wise linear algebra can scale linearly with N once the parameter and local dimensions are fixed.

Remark 8.3 (What treewidth does not prove). A tree decomposition specifies where elimination may occur. An exact dynamic program also needs a closed message language whose size and update time are controlled. For discrete variables of bounded domain, a table can have size exponential in the bag width and independent of the total number of bags. For real polynomial constraints, a projected semialgebraic set can require new polynomials, sign conditions, and components. The numerical dimension of a separator does not by itself bound this descriptive growth.

Remark 8.4 (Uniform and parameterized readings). The width bound supports a parameterized analysis in $P + K$. A claim of uniform polynomial time additionally requires polynomial dependence on $P + K$, not merely tractability for each fixed width. The cubic linear-algebra bound later in the paper has that dependence for evaluated matrices; the exact semialgebraic message problem does not yet have such a bound.

9 Cotangent Message Passing

This section states the exact theorem proved by the current CMP construction. It is a separator theorem for a linearized row space.

Let L_A, L_B, X_A, X_S, X_B be vector spaces over a field k , and let

$$J_A : L_A \rightarrow X_A, \quad J_{AS} : L_A \rightarrow X_S, \quad J_{BS} : L_B \rightarrow X_S, \quad J_B : L_B \rightarrow X_B \quad (44)$$

be linear maps. A pair $(\lambda, \mu) \in L_A \oplus L_B$ generates the block-angular row combination

$$(J_A \lambda, J_{AS} \lambda + J_{BS} \mu, J_B \mu). \quad (45)$$

Definition 9.1 (Intrinsic CMP message). *The message from the A-side to the separator is*

$$\boxed{\mathcal{E}(A \rightarrow S) := J_{AS}(\ker J_A) \subseteq X_S.} \quad (46)$$

Equivalently,

$$\mathcal{E}(A \rightarrow S) = \{J_{AS}\lambda : \lambda \in L_A, J_A\lambda = 0\}. \quad (47)$$

Theorem 9.2 (Linear separator factorization). *For $s \in X_S$ and $b \in X_B$, the following are equivalent:*

$$\exists \lambda \in L_A, \exists \mu \in L_B : J_A\lambda = 0, \quad s = J_{AS}\lambda + J_{BS}\mu, \quad b = J_B\mu; \quad (48)$$

$$\exists e \in \mathcal{E}(A \rightarrow S), \exists \mu \in L_B : s = e + J_{BS}\mu, \quad b = J_B\mu. \quad (49)$$

Proof. From [equation \(48\)](#), set $e = J_{AS}\lambda$; the condition $J_A\lambda = 0$ implies $e \in \mathcal{E}(A \rightarrow S)$, yielding [equation \(49\)](#). Conversely, membership $e \in \mathcal{E}(A \rightarrow S)$ supplies $\lambda \in \ker J_A$ with $e = J_{AS}\lambda$; substitution yields [equation \(48\)](#). $\square$

The theorem has no rank, smoothness, or invertibility hypothesis. It is exact for the stated linear maps.

9.1 Matrix form and basis independence

Suppose row vectors of $P_A^\perp$ form a basis of the relevant left null space. Then

$$M_{A \rightarrow S} = P_A^\perp J_{AS} \quad (50)$$

has row space $\mathcal{E}(A \rightarrow S)$. A different null-space basis left-multiplies $P_A^\perp$ by an invertible matrix and therefore gives the same message subspace. Redundant rows may be removed without altering that subspace.

Proposition 9.3 (Message dimension). *If X_S is finite-dimensional, then*

$$\dim_k \mathcal{E}(A \rightarrow S) \leq \dim_k X_S. \quad (51)$$

Proof. The message is a linear subspace of X_S . $\square$

9.2 Recursive linearized elimination

At a fixed point, each leaf sends a basis for the differential relations that survive elimination of its internal coordinates. At an internal node, child messages are stacked with the node's own rows, internal columns are eliminated, and the result is rank-reduced on the parent separator. Induction using [theorem 9.2](#) proves that the root message is exact for the global *linearized* row space.

For dense matrices of dimension at most $k = P + K$, conventional null-space and rank computations use $O(k^3)$ field operations per bag. With N bags, this gives

$$O(N(P + K)^3) \quad (52)$$

field operations for the evaluated linear problem. The coefficient field and the cost of representing its elements remain part of any bit-complexity claim.

9.3 Limit of the theorem

The message in [equation \(46\)](#) is homogeneous first-order data. It contains directions generated by linear combinations of derivatives. It does not, by definition, contain the constant terms of the input polynomials or a description of their real zero set. The next section tests whether it can nevertheless serve as a complete feasibility message and gives a negative answer.

10 Global exactness and decision completeness

An exact separator algorithm for EXACT-ReLU-NNTP-DECISION needs messages with zeroth-order semantics. If T_v is a rooted subtree, s its separator assignment, and x its internal variables, the required invariant is

$$s \in \text{Msg}_v \iff \exists x F_{T_v}(x, s). \quad (53)$$

Soundness is the left-to-right implication; completeness is the converse. At the root, nonemptiness of the message must be equivalent to global feasibility.

The semantic sets Q_i in [equation \(14\)](#) satisfy this invariant for leaf subtrees. Their conjunction satisfies it for the dataset. What is missing is a polynomially bounded representation and a polynomial-time projection/merge procedure.

10.1 A one-ReLU audit family

Fix a positive real target y . Consider one shared scalar parameter θ , local preactivation z , and local activation a , with constraints

$$z - \theta = 0, \quad a = \text{ReLU}(z), \quad a - y = 0. \quad (54)$$

Lemma 10.1 (Exact projection of the audit family). *For $y > 0$, the exact surviving parameter set of [equation \(54\)](#) is $Q_y = \{y\}$.*

Proof. The affine equation gives $z = \theta$ and the target equation gives $a = y > 0$. The ReLU equation then forces $z = a = y$, so $\theta = y$. Conversely, $(\theta, z, a) = (y, y, y)$ satisfies every constraint. $\square$

Use the complementarity polynomial for the ReLU equation and set

$$f_1 = z - \theta, \quad f_2 = a(a - z), \quad f_3 = a - y. \quad (55)$$

At the satisfying point $p_y = (\theta, z, a) = (y, y, y)$, their Jacobian rows in the column order (θ, z, a) are

$$J_y = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -y & y \\ 0 & 0 & 1 \end{bmatrix}. \quad (56)$$

For a row combination $(\lambda_1, \lambda_2, \lambda_3)$, eliminating the local (z, a) columns imposes

$$\lambda_1 - y\lambda_2 = 0, \quad y\lambda_2 + \lambda_3 = 0. \quad (57)$$

The surviving parameter coefficient is $-\lambda_1$.

Theorem 10.2 (Scalar cotangent message loses the target). *For every $y \neq 0$, the message*

$$E_y := \{-\lambda_1 : \lambda_1 - y\lambda_2 = 0, \quad y\lambda_2 + \lambda_3 = 0\} \quad (58)$$

equals $\mathbb{R}$.

Proof. Let $s \in \mathbb{R}$. Choose $\lambda_1 = -s$, $\lambda_2 = -s/y$, and $\lambda_3 = s$. These choices satisfy [equation \(57\)](#) and give $-\lambda_1 = s$. Hence $\mathbb{R} \subseteq E_y$, while the reverse inclusion is immediate. $\square$

Corollary 10.3 (No exact decoder from the scalar message alone). *There is no function*

$$D : \{\text{subsets of } \mathbb{R}\} \times \mathbb{R} \rightarrow \{\text{false}, \text{true}\} \quad (59)$$

such that, for every $y > 0$ and $\theta \in \mathbb{R}$,

$$D(E_y, \theta) = \text{true} \iff \theta \in Q_y. \quad (60)$$

Proof. By [theorem 10.2](#), $E_1 = E_2 = \mathbb{R}$. By [lemma 10.1](#), $1 \in Q_1$ and $1 \notin Q_2$. Thus the same input pair $(\mathbb{R}, 1)$ would have to receive both Boolean values. $\square$

This counterexample occurs at $y > 0$, where the ReLU is on its active linear branch. The failure is not caused by the switching boundary.

10.2 The symbolic version of the obstruction

Even before evaluation, formal differentiation erases the target constant:

$$\frac{\partial}{\partial x_j}(a - y) = \frac{\partial}{\partial x_j}(a - y') \quad \text{for every variable } x_j. \quad (61)$$

Therefore the symbolic Jacobian row family of the audit architecture is independent of y in its target row. Yet two positive-target samples sharing θ are jointly feasible exactly when their targets agree. In particular, targets $(1, 1)$ form a yes-instance and targets $(1, 2)$ form a no-instance with the same first-order target rows.

Theorem 10.4 (Cotangent-only global exactness is false). *The intrinsic first-order message $J_{AS}(\ker J_A)$, whether evaluated at the positive satisfying point or formed solely from the symbolic Jacobian rows, is not a complete invariant of exact parameter projection for all instances of EXACT-RELU-NNTP-DECISION.*

Proof. The evaluated claim follows from [corollary 10.3](#). The symbolic claim follows because the family of derivatives of the target polynomial is independent of its constant, whereas exact feasibility in [equation \(54\)](#) depends on that constant. $\square$

10.3 What remains possible

[Theorem 10.4](#) does not rule out every algorithm that uses the original equations together with cotangent summaries. It proves that any correct algorithm must preserve additional zeroth-order information. The tautologically complete message

$$\text{ZMsg}_v := \{s : \exists x_{\text{int}(T_v)} F_{T_v}(x, s)\} \quad (62)$$

satisfies [equation \(53\)](#). Computing [equation \(62\)](#) is existential projection, however, and general real quantifier elimination can create a much larger representation [[Col75](#), [Ren92](#), [BPR06](#)].

The unresolved repair theorem is therefore precise: find a representation class containing every message in [equation \(62\)](#) for the stated ReLU training systems, prove closure under leaf construction, conjunction, and projection, and bound both representation length and exact Turing running time by a polynomial in the original input length. No such theorem is established by the present first-order framework.

11 CMP decision algorithms: proved and required forms

Two procedures must be distinguished. The first is an exact algorithm for an evaluated linearized system. The second is the desired but currently unconstructed exact feasibility decider.

11.1 Differential CMP

Given numerical Jacobian blocks at a specified point p , differential CMP performs the following operations.

1. For each sample, form $C_i(p)$ and $D_i(p)$.
2. Compute a basis $P_i^\perp$ for $\ker D_i(p)^\top$.
3. Form the local parameter message

$$M_i = P_i^\perp C_i(p). \quad (63)$$

4. Stack the messages and remove linearly dependent rows.
5. Accept a proposed parameter direction $\delta\theta$ if and only if the root matrix annihilates $\delta\theta$.

Theorem 11.1 (Correctness of differential CMP). *At a specified real point p , the root test accepts $\delta\theta$ if and only if there exist local directions $\delta S^{(i)}$ satisfying all linearized equations*

$$C_i(p)\delta\theta + D_i(p)\delta S^{(i)} = 0. \quad (64)$$

Proof. Apply [proposition 7.3](#) independently to every sample and conjoin the resulting conditions. Equivalently, recursively apply [theorem 9.2](#) along the star decomposition. $\square$

This procedure assumes the point p is already supplied. It neither tests whether p exists nor decides the original instance.

11.2 The required exact procedure

An exact decision algorithm would need a finite message representation R_v for every subtree with a denotation $\llbracket R_v \rrbracket \subseteq \mathbb{R}^{S_v}$. Its operations must satisfy:

Leaf construction

$\llbracket R_v \rrbracket$ is exactly the projection of the leaf's ReLU constraints to its separator.

Merge

Combining child representations denotes the intersection of their lifted feasible sets with the current bag constraints.

Contraction

Eliminating internal variables denotes existential projection, with both soundness and completeness.

Root test

The algorithm returns YES exactly when the root denotation is nonempty.

Complexity

Representation length and the time for every operation are polynomial in the original binary input length.

If arbitrary quantifier-free formulas over the reals are used as representations, the first four requirements follow from standard logical semantics. The fifth does not follow. If only the cotangent subspaces of [definition 9.1](#) are used, the fifth is inexpensive but the first and third fail by [theorem 10.4](#).

Open Problem 11.2 (Polynomial semantic message). *Construct a representation and algorithms satisfying all five requirements for every finite instance of EXACT-RELU-NNTP-DECISION, or prove that no such representation exists under an explicitly stated computational model.*

11.3 No hidden root oracle

The root test must be included in the complexity analysis. A procedure that propagates compact differential messages and then invokes an unrestricted semialgebraic feasibility solver at the root has not placed the language in P ; it has moved the original difficulty into the final operation. The same applies if a “message entry” is an unevaluated existential formula whose nonemptiness remains ETR-hard.

12 Time, space, and bit complexity

12.1 Arithmetic complexity of differential CMP

Let $K = \max_i K_i$ and $k = P + K$. Suppose every evaluated Jacobian block is an explicitly supplied matrix over a field and has dimensions $O(k)$. Classical dense elimination computes a null-space basis, multiplies the surviving blocks, and row-reduces the result using $O(k^3)$ field operations per bag. Therefore

$$T_{\text{arith}}(N, P, K) = O(N(P + K)^3). \quad (65)$$

This is a valid bound for [theorem 11.1](#).

If $N, P, K \leq n$, then

$$N(P + K)^3 \leq n(2n)^3 = 8n^4. \quad (66)$$

Thus the number of field operations is uniformly $O(n^4)$ once the matrices are present. For fixed P, K , [equation \(65\)](#) is linear in N .

12.2 Space

A streaming implementation that processes and discards leaf matrices uses

$$O(V + E + (P + K)^2) \quad (67)$$

field elements in addition to constant bookkeeping per bag, where $V + E$ denotes the explicitly stored constraint incidence structure. Retaining all messages uses

$$O(V + E + N(P + K)^2) \quad (68)$$

field elements.

12.3 Why arithmetic counting is not a P theorem

Membership in P is a bit-complexity statement. The following data must be specified and bounded:

1. the representation of every coefficient;
2. the bit length of intermediate matrix entries;
3. the cost of exact zero testing and rank decisions;

4. the representation of any algebraic real at which a Jacobian is evaluated; and
5. the size and construction cost of the zeroth-order message needed for global feasibility.

For an explicitly given rational matrix, fraction-free elimination controls coefficient growth and yields polynomial bit complexity in the matrix dimensions and input bit length [Bar68]. This fact does not close the present proof: a feasible point can have irrational coordinates and is not part of the input, while finding or deciding the existence of such a point is the original nonlinear problem.

Similarly, treating symbolic entries as elements of a multivariate quotient ring changes the operation model. Rank, kernels, and equality testing over that ring cannot be charged as unit-cost field operations. Their representations can require polynomial reduction, localization, or ideal membership computations.

Theorem 12.1 (Conditional bit-complexity implication). *Assume there is an exact CMP representation satisfying open Problem 11.2, with total bit running time*

$$N \text{ poly}(P + K, L) \tag{69}$$

and a polynomial-time root nonemptiness test. Then EXACT-RELU-NNTP-DECISION $\in P$.

Proof. The instance dimensions and coefficient lengths are bounded by its explicit binary encoding length. Under the assumed bound, the stated CMP procedure is a deterministic polynomial-time Turing decider whose acceptance semantics is exactly $\text{Feas}(I)$. $\square$

Remark 12.2 (Current status). The hypothesis of theorem 12.1 is **OPEN**. Equation equation (66) proves only the arithmetic inequality; it does not construct a global decider.

13 The ETR bridge

To obtain $\exists\mathbb{R}$ -hardness, a reduction must begin with an explicitly encoded ETR sentence

$$\varphi = \exists x_1, \dots, x_m : \Psi(x_1, \dots, x_m) \tag{70}$$

and construct in polynomial time an instance I_φ of the exact language in section 3 such that

$$\varphi \text{ is true} \iff \text{Feas}(I_\varphi). \tag{71}$$

Related neural-network training languages are known to be $\exists\mathbb{R}$ -complete [AKM21]. A citation to that result does not by itself prove equation (71): the architecture, activation function, loss/threshold convention, permitted trainable parameters, and exact acceptance predicate must match, or a polynomial model-conversion reduction must be given.

13.1 Required gadget ledger

The reduction must provide the following constructions with forward and reverse correctness proofs.

Object	Required invariant	Status
Rational constant	A designated signal/parameter equals the encoded rational.	Open
Signed variable	An unrestricted real variable is represented despite nonnegative ReLU outputs.	Open
Equality	Two represented expressions are equal iff the exact output constraints hold.	Open
Addition	A represented output equals the sum of represented inputs.	Open
Multiplication	A represented output equals the product of two represented unknowns, with legal sharing in the network model.	Open
Weak/strict sign	The source sign predicate is preserved using only permitted exact constraints.	Open
Conjunction	All source constraints hold simultaneously under one shared assignment.	Open
Reuse	Multiple occurrences of one source variable denote one real value without exponential copying.	Open

Unknown weights multiply activations in an affine layer, so the network equations contain bilinear terms. Turning that observation into a multiplication gadget requires more than noting the term’s existence: both operands must be represented as source variables, their signs must be handled, and all uses must remain synchronized by the permitted parameter-sharing rules.

13.2 Size and bit bounds

For a valid many-one reduction, the construction must satisfy

$$|\langle I_\varphi \rangle| \leq \text{poly}(|\langle \varphi \rangle|). \quad (72)$$

This includes the number of nodes, samples, trainable incidences, local state variables, and the bit lengths of rational constants. Correctness must be proved in both directions; a gadget that introduces unintended ReLU branches is insufficient.

Open Problem 13.1 (Model-matching ETR reduction). *Prove [equation \(71\)](#) for the exact target-equality language of [section 3](#), with polynomial construction size and coefficient bit length, or modify the language and every upstream theorem in a stated, consistent manner.*

Remark 13.2 (Current status). Membership in $\exists\mathbb{R}$ is proved in [proposition 4.4](#). Hardness for this exact formal language remains **OPEN** until [open Problem 13.1](#) is closed.

14 The conditional complexity collapse

The class-theoretic part of the proposed argument is short once its two substantive certificates exist.

Theorem 14.1 (Conditional $P = \exists\mathbb{R}$). *Assume:*

1. EXACT-RELU-NNTP-DECISION $\in P$; and
2. $\text{ETR} \leq_p \text{EXACT-RELU-NNTP-DECISION}$.

Then $P = \exists\mathbb{R}$.

Proof. The reduction and closure of P under polynomial-time many-one reductions imply $\text{ETR} \in P$, hence $\exists\mathbb{R} \subseteq P$. The standard containment $P \subseteq \exists\mathbb{R}$ gives equality. $\square$

Corollary 14.2 (Conditional $P = NP = \exists\mathbb{R}$). *Under the hypotheses of [theorem 14.1](#),*

$$P = NP = \exists\mathbb{R}. \quad (73)$$

Proof. Use $P \subseteq NP \subseteq \exists\mathbb{R} = P$. $\square$

Remark 14.3 (No unconditional collapse claim). The first hypothesis is not obtained from differential CMP because cotangent-only global exactness is false by [theorem 10.4](#). The second hypothesis requires the model-matching construction of [open Problem 13.1](#). Consequently, [theorem 14.1](#) and [corollary 14.2](#) are logical implications, not proofs that their hypotheses hold.

The Lean development accompanying this paper formalizes an abstract version of these implications by packaging the two assumptions as a CMP decision certificate and an ETR bridge certificate. It does not instantiate either certificate for the exact language.

15 Globality, singularities, and ReLU switching boundaries

The exact ReLU encoding covers the switching boundary semantically. At $z = a = 0$, however, the complementarity polynomial has vanishing gradient. Thus an argument that identifies cotangent rank with the codimension of a smooth real manifold cannot be extended to this point without additional work.

15.1 Strata and their intersections

For a fixed activation pattern, ReLU constraints reduce to affine equalities and weak sign conditions. The complete network is the union over activation patterns, with different strata meeting where one or more preactivations vanish. An exact decision procedure must account for all strata and their intersections. Enumerating patterns directly can be exponential in the number of gates.

Square-slack variables encode the weak inequalities without selecting a pattern, but they do not make the resulting variety smooth. At a zero slack, the derivative of u^2 also vanishes. Hence polynomialization preserves the exact feasible projection while potentially adding singular coordinates.

15.2 Three levels of information

The following objects must not be conflated:

1. the global module $\Omega_{\mathcal{R}/\mathbb{R}}$ over the coordinate ring;
2. its fiber, or an evaluated Jacobian row space, at a chosen real point;
3. the zeroth-order semialgebraic projection of the real solution set.

The tensor and Kähler decompositions concern the first object. Differential CMP operates on the second. Exact-NNTP feasibility is determined by the third.

15.3 The obstruction is stronger than a boundary problem

Although boundary completeness remains necessary for any all-ReLU theorem, the audit family in [section 10](#) lies on the active branch with $y > 0$. Its failure shows that a smooth-locus extension cannot by itself upgrade first-order CMP to a zeroth-order decider. Both issues must be solved: the message must retain exact feasible-set information, and it must handle singular switching strata.

16 Related work

The existential theory of the reals. ETR is the nonemptiness problem for semialgebraic sets presented by existential formulas. Its role as a complexity class, known complete problems, and the containments between NP and PSPACE are surveyed by Schaefer, Cardinal, and Miltzow [\[SCM24\]](#). General decision and quantifier elimination algorithms include cylindrical algebraic decomposition [\[Col75\]](#), the PSPACE method of Canny [\[Can88\]](#), and the complexity analyses of Renegar [\[Ren92\]](#); see also [\[BPR06\]](#).

Exact neural network training and verification. Abrahamsen, Kleist, and Miltzow proved $\exists\mathbb{R}$ -completeness for a neural network training problem [\[AKM21\]](#). Their result makes precise why a claimed polynomial-time exact training decider requires an equally precise comparison of formal languages. ReLU network verification has also motivated specialized satisfiability procedures such as Reluplex [\[KBD⁺17\]](#). Verification of a fixed network and existential training of its weights are related semialgebraic problems but have different quantified variables and should not be identified without a reduction.

Kähler differentials. Kähler differentials provide a functorial algebraic representation of derivations, conormal relations, and tangent/cotangent information. The Jacobian presentation, base change, tensor-product formula, and transitivity sequence used here are standard commutative algebra [\[Eis95, The\]](#). Their availability over singular rings does not make a first-order invariant a complete description of real points.

Treewidth and elimination. Tree decompositions expose small interfaces between otherwise separated subproblems [\[Bod96\]](#). Exact message passing is efficient when the message domain has a bounded representation closed under local combination and elimination. In this paper the star decomposition is elementary; the research question is the representation complexity of real semialgebraic separator messages.

Exact linear algebra. The arithmetic-operation count for CMP uses conventional dense elimination. Fraction-free methods such as Bareiss elimination control intermediate integer growth for exact rational matrices [\[Bar68\]](#). These methods do not replace nonlinear real projection or construct a feasible evaluation point.

17 Discussion and repair program

The structural decomposition remains useful despite the failed completeness claim. It identifies exactly where samples interact and provides a compact description of infinitesimal compatibility. The counterexample identifies the minimum semantic information that any exact extension must add.

17.1 Potential message enrichments

The following are research directions rather than established polynomial-time repairs.

Semialgebraic projection messages. Using the exact set in [equation \(62\)](#) gives perfect semantics. The unresolved issue is whether this special family has polynomial descriptive complexity and supports polynomial-time elimination. General quantifier elimination does not supply the desired bound.

Affine or homogenized first-order messages. Augmenting a derivative matrix with constant terms prevents the immediate loss of y in the audit family and is exact for affine constraint systems. For nonlinear systems, an affine row space still does not generally determine the zero set or its projection.

Higher jets. Higher-order derivative data can retain more of a bounded-degree polynomial. A successful method would need a uniform order, a basis-independent merge operation, correct treatment of inequalities and real components, and a polynomial representation bound. None follows from the first-order separator theorem.

Ideals, real radicals, and elimination ideals. An algebraic message could retain an elimination ideal rather than a cotangent subspace. Ordinary elimination ideals describe Zariski closures of complex projections; exact real feasibility can additionally require inequalities and real-radical information. Gröbner or quantifier-elimination representations may grow substantially. The special block structure would have to yield a new bound.

17.2 Priority order for the target theorem

Further work should proceed in the following dependency order:

1. propose a zeroth-order message representation that distinguishes the audit family;
2. prove its exact semantic invariant for leaf, merge, contraction, and root operations;
3. attack it with small disconnected, singular, and incompatible-target instances;
4. prove a polynomial representation-size theorem;
5. derive a Turing bit-complexity bound;
6. formalize the model-matching ETR reduction; and
7. only then instantiate the conditional collapse theorem.

This ordering prevents an arithmetic speedup for an auxiliary linear problem from being mistaken for a complexity classification of the nonlinear decision language.

17.3 A publishable unconditional core

Independently of the open collapse target, the following package is unconditional: exact ReLU polynomialization including the boundary; the shared-base tensor decomposition; the relative Kähler decomposition; the block-angular Jacobian; the intrinsic linear CMP factorization; and the in-scope impossibility theorem for cotangent-only exact decoding. A future positive algorithm can build on this audited base.

18 Conclusion

Exact ReLU training has a strong sample-separable algebraic structure. Its coordinate ring is assembled from sample-local factors over one shared parameter algebra, its relative Kähler differentials decompose after base change, and its Jacobian is block-angular. Cotangent Message Passing exploits this structure to eliminate local coordinates exactly in the linearized row space, using messages whose vector-space dimension is bounded by the separator.

Exact feasibility requires more. The one-ReLU audit family proves that the existing first-order message can be identical for problems with different zeroth-order parameter projections. Accordingly, the present results do not establish a polynomial-time decider for EXACT-RELU-NNTP-DECISION, and therefore do not establish $P = \exists\mathbb{R}$ or $P = NP$. The missing theorem is now explicit: a polynomially represented separator message whose denotation is the exact real existential projection at every subtree, together with polynomial Turing-model operations and a model-matching ETR reduction. Resolving that theorem—or proving a barrier for the proposed message families—is the next mathematical task.

A Full exact-ReLU algebra

This appendix expands the proof obligations behind [lemmas 4.1](#) and [4.2](#).

A.1 Case partition

The equation $a(a - z) = 0$ gives the algebraic alternatives $a = 0$ and $a = z$. The two weak inequalities select the valid half of each branch:

Algebraic branch	Consequence of inequalities	ReLU regime	Included boundary
$a = 0$	$z \leq 0$	inactive	$z = 0$
$a = z$	$z \geq 0$	active	$z = 0$

At $z = a = 0$, both algebraic branches meet, but they represent the same ReLU assignment. The complementarity description therefore has no spurious (z, a) -point.

A.2 Square-slack fibers

If $a > 0$, the equation $a = u^2$ has two real slack witnesses. If $a = 0$, it has the unique witness $u = 0$. The same holds for $a - z = v^2$. Hence the projection of the slack variety to (z, a) is exactly the ReLU graph, although the map is generally many-to-one.

For one gate, the equality-only encoding adds two variables and three equalities, all of degree at most two. For G ReLU gates and N samples, the construction adds $2GN$ real variables and $3GN$ polynomial equalities. With an explicit architecture and dataset, this is polynomial in the input length.

A.3 Boundary Jacobian

For

$$h_1 = a - u^2, \tag{74}$$

$$h_2 = a - z - v^2, \tag{75}$$

$$h_3 = a(a - z), \quad (76)$$

the Jacobian in (z, a, u, v) at the boundary origin is

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (77)$$

The complementarity row vanishes and the slack derivatives vanish. Thus existential square polynomialization provides exact semantics but does not remove singular behavior from the equality presentation.

A.4 Formal verification correspondence

The Lean theorem `exactReLU_iff_complementarity` proves [lemma 4.1](#). The theorem `exactReLU_iff_polynomialized` proves [lemma 4.2](#). Separate lemmas cover active, inactive, and boundary regimes. These results concern the scalar gate and are composed sample-by-sample in the mathematical encoding.

B Tensor products over the shared parameter algebra

We give a more explicit two-factor proof of [theorem 5.2](#). Let $B_1 = A[X]$, $B_2 = A[Y]$ for disjoint finite variable sets and let $I_1 \triangleleft B_1$, $I_2 \triangleleft B_2$. The canonical isomorphism

$$\alpha : B_1 \otimes_A B_2 \xrightarrow{\sim} A[X, Y] \quad (78)$$

sends $f(X) \otimes g(Y)$ to $f(X)g(Y)$. Let J be the ideal of the tensor product generated by the images of $I_1 \otimes B_2$ and $B_1 \otimes I_2$. Right exactness of tensor product gives canonical surjections whose composite induces

$$(B_1/I_1) \otimes_A (B_2/I_2) \xrightarrow{\sim} (B_1 \otimes_A B_2)/J. \quad (79)$$

Under α , the ideal J maps to $I_1 A[X, Y] + I_2 A[X, Y]$. Therefore

$$\frac{A[X, Y]}{I_1 A[X, Y] + I_2 A[X, Y]} \cong (B_1/I_1) \otimes_A (B_2/I_2). \quad (80)$$

Associativity of tensor products supplies the finite iterated statement. No ordering of samples is mathematically preferred; different parenthesizations are related by the canonical associators.

B.1 Why base choice matters

If one instead forms a tensor product over k , then the two copies of $A = k[\theta]$ remain independent. For example,

$$k[\theta] \otimes_k k[\theta] \cong k[\theta^{(1)}, \theta^{(2)}], \quad (81)$$

whereas

$$k[\theta] \otimes_{k[\theta]} k[\theta] \cong k[\theta]. \quad (82)$$

The latter is the correct model for a single shared trainable parameter vector.

B.2 Real points do not commute with arbitrary projection claims

The tensor identity is an identity of algebras. It implies that the global equations contain no hidden cross-sample polynomial except through the shared base. It does not imply

$$\text{Proj}_\theta V(I) = V(I \cap A) \quad (83)$$

as equality of real sets without closure and real-algebraic qualifications. This is why the semantic projection in [proposition 3.4](#) is kept separate from elimination ideals.

C Kähler exact sequences and base change

For completeness, we record the universal-property maps used in [theorem 6.1](#). Given A -algebras B, C and $D = B \otimes_A C$, define

$$\delta : D \rightarrow (\Omega_{B/A} \otimes_B D) \oplus (\Omega_{C/A} \otimes_C D) \quad (84)$$

on pure tensors by

$$\delta(b \otimes c) = (db \otimes (1 \otimes c), dc \otimes (b \otimes 1)). \quad (85)$$

This is an A -derivation. Conversely, restricting the universal derivation of D along $B \rightarrow D$ and $C \rightarrow D$ gives maps from the two summands to $\Omega_{D/A}$. The maps are inverse because they agree with the universal derivations on pure tensors.

Applied iteratively to $D = \mathcal{R}$, this yields [equation \(31\)](#). The tensor factors on the right are essential: $\Omega_{\mathcal{R}^{(i)}/A}$ is naturally a module over the local ring, while the direct sum must be compared as a module over the global ring.

C.1 Presentation for a quotient

If $P = k[x_1, \dots, x_n]$, $I = (f_1, \dots, f_m)$, and $B = P/I$, the conormal sequence is

$$I/I^2 \longrightarrow \Omega_{P/k} \otimes_P B \longrightarrow \Omega_{B/k} \longrightarrow 0. \quad (86)$$

Under the basis $dx_1, \dots, dx_n$, the first map sends the class of f_i to

$$df_i = \sum_j \frac{\partial f_i}{\partial x_j} dx_j. \quad (87)$$

Choosing the displayed generators of I therefore gives the Jacobian presentation [equation \(8\)](#). The presentation need not be minimal, and the first arrow need not be injective.

C.2 Evaluation

For a real point $p : B \rightarrow \mathbb{R}$, tensoring with $\mathbb{R}$ through p gives

$$\mathbb{R}^m \xrightarrow{J_F(p)^\top} \mathbb{R}^n \longrightarrow \Omega_{B/k} \otimes_{B,p} \mathbb{R} \longrightarrow 0. \quad (88)$$

This operation retains the derivatives evaluated at p and discards polynomial coefficients that vanish under evaluation. It cannot serve as a procedure for deciding whether any real point p exists.

D Full separator factorization proof

Define the global linear map

$$T : L_A \oplus L_B \rightarrow X_A \oplus X_S \oplus X_B \quad (89)$$

by

$$T(\lambda, \mu) = (J_A \lambda, J_{AS} \lambda + J_{BS} \mu, J_B \mu). \quad (90)$$

We wish to eliminate the X_A -coordinate from $\text{im } T$. Its exact restriction to zero in that coordinate is

$$\mathcal{Z} := \{(s, b) : (0, s, b) \in \text{im } T\} \quad (91)$$

$$= \{(J_{AS} \lambda + J_{BS} \mu, J_B \mu) : J_A \lambda = 0\}. \quad (92)$$

Since $J_{AS} \lambda$ ranges exactly over $\mathcal{E}(A \rightarrow S) = J_{AS}(\ker J_A)$, we obtain

$$\mathcal{Z} = \{(e + J_{BS} \mu, J_B \mu) : e \in \mathcal{E}(A \rightarrow S), \mu \in L_B\}. \quad (93)$$

This is [theorem 9.2](#) as an equality of sets.

D.1 Uniqueness at the level of subspaces

The construction $J_{AS}(\ker J_A)$ is canonical for the given pair of linear maps. A matrix $P_A^\perp$ is only a choice of generators. If P and P' are full-row-rank matrices with the same row space, there is an invertible change-of-basis matrix U with $P' = UP$, and hence

$$\text{rowspan}(P' J_{AS}) = \text{rowspan}(UP J_{AS}) = \text{rowspan}(P J_{AS}). \quad (94)$$

If dependent rows are allowed, equality of the image subspace still follows from the intrinsic definition.

D.2 Recursive invariant for linear systems

For a rooted tree of block-linear systems, let L_v be the row-coefficient space in the subtree and let X_{S_v} be the parent-separator coordinate space. The message is the image on X_{S_v} of those row combinations whose coordinates internal to the subtree vanish. The leaf case is [definition 9.1](#); the induction step is [theorem 9.2](#) after stacking child messages with the current rows. At the root, the message therefore gives exactly the relations in the global linear row space after all nonroot coordinates have been eliminated.

This invariant concerns generated linear relations. Replacing “row combination whose internal coordinates vanish” by “assignment extendable to a nonlinear real solution” changes the semantic object and invalidates this proof.

E Global CMP exactness audit

This appendix spells out the logical incompatibility behind [corollary 10.3](#).

For positive y , let

$$\text{One}(y, \theta) :\iff \exists z, a : z = \theta \wedge a = \text{ReLU}(z) \wedge a = y. \quad (95)$$

Then

$$\text{One}(y, \theta) \iff \theta = y. \quad (96)$$

Let E_y be the scalar message in [theorem 10.2](#). Assume for contradiction that a decoder D satisfies

$$D(E_y, \theta) \iff \text{One}(y, \theta) \quad (y > 0). \quad (97)$$

At $(y, \theta) = (1, 1)$, the right side is true, so $D(E_1, 1)$ is true. Since $E_1 = E_2 = \mathbb{R}$, this implies $D(E_2, 1)$ is true. But [equation \(96\)](#) makes $\text{One}(2, 1)$ false, a contradiction.

E.1 Two-sample Boolean separation

Define

$$\text{Two}(y_1, y_2) : \iff \exists \theta : \text{One}(y_1, \theta) \wedge \text{One}(y_2, \theta). \quad (98)$$

For positive targets,

$$\text{Two}(y_1, y_2) \iff y_1 = y_2. \quad (99)$$

Thus $\text{Two}(1, 1)$ is true and $\text{Two}(1, 2)$ is false. The formal derivative of the target equation $a - y$ is the same for both target values. A message construction that depends only on those derivatives cannot distinguish the Boolean answers.

E.2 Scope of the impossibility theorem

The theorem rules out a decoder whose only semantic summary of an eliminated subproblem is the cotangent message subspace. It does not rule out carrying the original target constants, nonlinear equations, sign data, elimination ideals, or complete semialgebraic projections. Each enrichment must be analyzed for semantic sufficiency and polynomial representation size.

E.3 Lean correspondence

The formal development contains:

- `scalarCMPMessage_eq_univ`;
- `scalarCMPMessage_one_eq_two`;
- `exact_projection_one_diff_two`;
- `no_exact_decoder_from_scalarCMPMessage_alone`; and
- `first_order_data_not_complete_for_exact_relu_decision`.

These theorems formalize the exact in-scope obstruction rather than a generic counterexample about unrelated varieties.

F Complexity ledger

Let a dense local matrix have $O(k)$ rows and columns. Conventional Gaussian elimination uses $O(k^3)$ arithmetic operations and $O(k^2)$ stored field elements. Matrix multiplication and rank reduction have the same classical cubic upper bound. Over N sample bags this yields $O(Nk^3)$ operations and, when streamed, $O(k^2)$ working field elements.

For $k = P + K$ and $N, P, K \leq n$, monotonicity gives

$$N(P + K)^3 \leq n(n + n)^3 \quad (100)$$

$$= n(2n)^3 \quad (101)$$

$$= 8n^4. \quad (102)$$

This inequality has been machine-checked in the Lean development.

F.1 Conditional rational-matrix bit bound

Suppose an $r \times c$ matrix with rational entries of bit length at most L is explicitly given. Clearing denominators produces an integer matrix with polynomially related bit length. Fraction-free elimination represents intermediate entries through minors; determinant bounds then give polynomial bit lengths in r, c, L . This supports a polynomial Turing bound for the linear-algebra phase [Bar68].

The premises are not automatically satisfied by symbolic CMP:

- a satisfying point is not given;
- its coordinates need not be rational;
- symbolic entries lie in a polynomial or quotient ring; and
- the zeroth-order message may contain several semialgebraic components.

F.2 Checklist for a future bit theorem

A complete proof of [equation \(69\)](#) must state:

1. a canonical binary encoding for every message;
2. polynomial bounds on message length after every merge and projection;
3. exact algorithms for equality, sign, and emptiness in that encoding;
4. coefficient-height bounds under every operation;
5. the number of operations along the tree; and
6. the cost of the root acceptance test.

Without these items, an $O(n^4)$ count of abstract ring or field operations is not a proof of membership in P .

G ETR reduction obligations

This appendix records a proof template for [open Problem 13.1](#). Normalize the source formula to a conjunction of primitive constraints only after proving that the normalization has polynomial size and preserves satisfiability. A suitable primitive source language might contain

$$x_i = c, \quad x_i + x_j = x_k, \quad x_i x_j = x_k, \quad x_i \geq 0, \quad (103)$$

but the completeness and normalization theorem for the chosen language must be cited or proved.

For each primitive constraint, a network gadget must come with a relation R_G between its interface values and satisfy

$$R_G(\text{interface}) \iff \text{source primitive holds.} \quad (104)$$

The reverse implication is essential: additional trainable weights or activation branches must not create spurious satisfying assignments.

G.1 Composition requirements

Gadgets must compose while preserving one value for every repeated source variable. If source variables are represented by shared trainable parameters, the reduction must show that all required occurrences of a parameter are legal under the architecture encoding. If they are represented by activations, the reduction must supply equality/copy gadgets and handle negative values.

The final construction must use exact target equalities as specified in [section 3](#); importing a theorem for total squared error below a threshold requires an explicit conversion. Likewise, a theorem for a different activation or for trainable architecture cannot be cited as hardness for this language without a reduction.

G.2 Audit table for every gadget

Audit item	Required proof
Forward correctness	Every source assignment extends to a network witness.
Reverse correctness	Every network witness restricts to a valid source assignment.
Boundary behavior	Zero preactivations introduce no extra source solutions.
Size	Nodes, edges, samples, and states grow polynomially.
Height	All rational coefficients have polynomial bit length.
Uniformity	A Turing machine constructs the encoded instance in polynomial time.
Scope	The output is exactly an instance of EXACT-RELU-NNTP-DECISION.

Only after every row is discharged may the manuscript state

$$\text{ETR} \leq_p \text{EXACT-RELU-NNTP-DECISION} \quad (105)$$

as an unconditional theorem.

H Lean 4 verification notes

The accompanying Lean 4 development is on the isolated repository branch `cmp-lean-verification`. This manuscript was initialized from commit `fcca060331defca7226db3e0aecc8bc3c93fcbdf`. GitHub Actions run 74 for that commit completed successfully. The workflow runs `lake build` and then rejects occurrences of `sorry`, `admit`, or a line beginning with `axiom` in the `CMPLean` directory.

Lean file	Verified content used by this manuscript
<code>ReLU.lean</code>	Complementarity, square-slack polynomialization, and active/inactive/boundary cases.
<code>DecisionProblem.lean</code>	Abstract Boolean existential feasibility semantics with shared parameters.
<code>ExactProjection.lean</code>	Feasibility iff existence in the shared-parameter projection.
<code>KahlerCore.lean</code>	Transitivity exactness and relative-map surjectivity from Mathlib.
<code>BlockPolynomial.lean</code>	Cross-sample formal partial derivatives vanish.
<code>AmbientTensor.lean</code>	Ambient shared-base polynomial tensor equivalence.
<code>Separator.lean</code>	Basis-free linear separator factorization.
<code>MessageSubmodule.lean</code>	Message as image of a kernel and separator-dimension bound.
<code>ComplexityArithmetic.lean</code>	The inequality $N(P + K)^3 \leq 8n^4$ under dimension bounds.
<code>ReLUDecisionTests.lean</code>	One- and two-sample positive-target feasibility.
<code>MessageCompleteness</code>	Identical scalar CMP messages with different exact projections;
<code>Audit.lean</code>	no message-only decoder.
<code>ZeroOrderAdversarial.lean</code>	Target constants vanish from first derivatives while Boolean feasibility changes.
<code>CollapseLogic.lean</code>	Conditional class-collapse logic from abstract decision and reduction certificates.

The formalization does not construct the CMP decision certificate or the ETR bridge certificate. In particular, the theorem named `P_eq_NP_eq_ExistsR` accepts those certificates as hypotheses. Its successful compilation verifies the final logical implication, not the missing global exactness or reduction premises.

The current formal audit is unusually informative because it includes a negative theorem internal to the exact ReLU scope. Any repaired CMP definition should first be instantiated on the one-ReLU and incompatible-target tests before a new global certificate is attempted.

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